ORIGINAL ARTICLE

# Mindfulness influences the psycho-social dimension of chronic pain: A randomized controlled clinical trial in Indian context

**Anirban Pal[1], Purnava Mukhopadhyay[2], Rajasree Biswas[3], Dipasri Bhattacharya[4]**

[1]Department of Anaesthesia and Pain Medicine, KPC Medical College and Hospital, Kolkata, West Bengal, India, [2]Department of Public Health, Kalyani ESI Hospital, Kalyani, West Bengal, India, [3]Department of Anaesthesia, Raigunj Medical College, Raiganj, West Bengal, India, [4]Department of Anaesthesia, R G Kar Medical College, Kolkata, West Bengal, India

## ABSTRACT

**Background and Aims:** In India, the awareness about the psycho-social dimension of chronic pain is minimal among physicians and patients. The research with community-based group therapies (like mindfulness) to address the psycho-social aspects in chronic pain patients remains limited. The aim of this randomized controlled trial was to see the effects of mindfulness on pain intensity, pain catastrophizing, chronic pain acceptance, perceived stress, well-being, and mindfulness characteristics.

**Materials and Methods:** In this two-site, parallel group, clinical trial, 170 patients attending pain outdoors of two government hospitals in West Bengal, India, were randomized to attend five weekly in-person mindfulness sessions (cases) or usual care sessions (controls) within the hospital premises. Pre-program and post-program data were collected and analyzed using statistical methods like repeated measures analysis of variance.

**Results:** In participants of the mindfulness group, significant changes post session were noted in pain intensity [F(1,326) = 15.0122; $P$ = 0.0001291], pain acceptance [F(1,326) = 4.5311; $P$ = 0.03403], and perceived stress score [F(1,326) = 13.2788; $P$ = 0.0003122] compared to pre-session. The changes in pain catastrophizing, World Health Organization well-being and Freiburg mindfulness inventory scores were non-specific.

**Conclusion:** Mindfulness had a positive influence on pain intensity, pain acceptance, and perceived stress of Indian chronic pain patients. The effects on pain catastrophizing, mindfulness characteristics, and well-being (non-specific) were also encouraging. Further studies will be required to substantiate these results.

**Key words:** Mindfulness, pain acceptance, pain catastrophizing, pain intensity, psycho-social

## INTRODUCTION

Chronic pain is emerging as a major health care problem, and 19.3% of the Indian population or 180–200 billion Indians are known to be suffering.[1] The belief that chronic pain results only from tissue damage (biomedical

**Address for correspondence:** Dr. Anirban Pal,
43/6/5 Jheel Road, Kolkata - 700 031, West Bengal, India.
E-mail: pal.anirban1@gmail.com

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model) is still prevalent in this part of the world. But chronic pain is a complex phenomenon that includes psychological and social components in addition to biological components (bio-psycho-social model).[2] Neither the psycho-social factors are routinely assessed in hospitals/clinics nor are they taken into account to improve treatment outcomes. The suffering patient need to be looked upon as a total human being, living with chronic pain in a challenging social world. The pain conditions present with a wide range of associated feelings/emotions and manifest various levels of psycho-social distress, reciprocally influencing one another and resulting in chronic pain conditions.[3] Physical and psychological adaptation to chronic pain does not constitute a psycho-pathology; rather, it reflects normal processes of the human condition.[4] Pain beliefs like pain catastrophizing (an exaggerated, negative interpretation of pain), fear of avoidance of work (a belief that all activities should be avoided to reduce pain), and low expectations for recovery put patients at a poor prognosis.[5]

Mindfulness is a mind–body intervention for chronic pain patients, typically delivered as group therapy in both clinical and community settings. In mindfulness training, one purposely brings one's attention to present-moment experiences, including thoughts, emotions, or sensations that arise, noticing them and letting them go without judgment.[6] In western countries, mindfulness started as mindfulness-based stress reduction (MBSR) programs and later gave way to abbreviated versions of mindfulness-based intervention (MBI) which were found to be equally effective.[7] Mindfulness-related pain relief and other effects were known to be associated with unique appraisal cognitive processes and share a common final pathway with other cognitive techniques.[8] For the past few decades, mindfulness has shown to effect the psychological health and quality of life of chronic pain patients in the West.[9]

In Indian context, the awareness about the psycho-social dimension of chronic pain remains limited, restricting the application of interventions like mindfulness in these patients. The present investigators conducted a pilot study and found mindfulness may have potential in Indian chronic pain patients also.[10] India differs in socio-culture characteristics from the West, and due to the cultural and genetic variation in pain causation, manifestation, and reporting, the Western studies cannot be extrapolated to the Indian population.[1] As chronic pain is emerging as a massive public health problem in India, the investigators search for a group intervention (in the form of mindfulness), which is evidence-based and socially acceptable and can be practically implemented at the ground level for the masses. A well-planned randomized controlled trial with mindfulness on Indian chronic pain patients was the need of the hour.

In this clinical trial, the investigators wanted to see the effects of a brief face-to-face mindfulness-based intervention on the different dimensions of chronic pain in a clinical setting. Mindfulness was used in combination with traditional medical treatment (without altering existing treatment regimens). This study focuses on addressing sub-clinical psycho-social associations but not treating those with psychiatric co-morbidities. The objective of this study was to see the effects on pain intensity [measured by numerical rating scale (NRS)], pain catastrophizing [measured by pain catastrophizing scale (PCS)], chronic pain acceptance [measured by chronic pain acceptance scale (CPAQ-8)], stress [measured by perceived stress scale (PSS-4)], well-being (measured by WHO-5-Well-being Index), and mindfulness characteristics [measured by Freiburg Mindfulness Inventory (FMI)]. We form a hypothesis that mindfulness will have a positive impact on different variables of Indian chronic pain patients, compared to the control group.

## MATERIALS AND METHODS

### Settings and design

This was a two-site, parallel group, randomized controlled trial where patients were recruited from outdoors of two government hospitals in West Bengal, India. The design adopted was uniform, the same research question/protocol was pursued in both institutions, and data collected were pooled for analysis. A mindfulness center collaborated with the hospitals to conduct in-person sessions in the hospital premises itself. The study period was June 2022 to December 2022. An institutional ethical clearance was obtained. The trial was registered with the clinical trial registry of India.

### Sample size calculation

The sample size calculation was made on the assumption of a standard deviation (SD) of 1.68 and 1.25 for the expected difference in pain intensity measured by NRS from the limited available previous studies by de Boer MJ *et al.* and Omidi A *et al.*, respectively. With a power of >80% to detect the above-mentioned difference using repeated measures analysis of variance (ANOVA) with a Type I error ($\alpha$) of <5%, the calculated minimum sample size was 36 in each group. Since the study was carried out in two centers, the minimum sample size was 72 in each group. So, the total sample size was 144. Hence, we started the study with 170 patients {taking into account attrition of at least 15% [144+ (144×.15) =166]}, who were randomly allocated into group M and group U in our two centers. A total of 165 participants (84 in group M and 81 in group U) completed the study.

### Participant selection and randomization

Chronic pain patients attending hospital outdoors were informed about the study. Initially, 200 patients were assessed for eligibility. The patients were sent to the psychiatric outdoor to eliminate any psychiatric illness before recruitment (if found excluded from study). The details of

patient recruitment, randomization, and post-treatment follow-up are depicted in Figure 1 (consort flow diagram).

### Inclusion criteria

1. The patient has an age of 18 to 65 years. 2. The patient suffers from a chronic non-cancer pain for more than 3 months. 3. The patient has a definitive provisional diagnosis of the chronic pain condition. 4. The patient has an intensity of pain >4 in NRS. 5. The patient is on a stable treatment regimen with no plans to change in the next 2 months. 6. The patient has attended three sessions of the five-session program. 7. The patient has education at least up to VIth standard. 8. The patient is a resident of India.

### Exclusion criteria

1. The patient has previous experience with mindfulness. 2. The patient has a major physical disability or problem in cognition/communication. 3. The patient undergoes any psychological therapy or psychiatric treatment at present.

### Randomization

Random selection of participants in each center to either group M or group U was done using a random sequence of numbers. The sequence was produced through the SPSS software's random selection procedure. Informed written consent was taken from all the patients before randomization. The participants were blinded to the type of intervention (mindfulness versus usual care) allocated. The medical treatment regimen of the patients was not interfered with and the intervention 'mindfulness' or 'usual care' was used additionally.

### Intervention

A five-session version of the original MBSR program was chosen for the study [detailed in Table 1]. The reason behind choosing this program was our previous experience with Indian chronic pain sufferers.[10] The chances of high dropouts and practical ease to conduct a brief format in hospital premises prompted us to adopt this MBI. Our focus was to retain maximum participants with minimal dropouts. This MBI had a face-to-face session with around 40 participants in a group, with approximately 1.5 hour durations, interactions, and a daily home practice of 20 minutes. A daily practice log was maintained by participants and verbally confirmed by study coordinators every week. The mindfulness instructor was trained via the

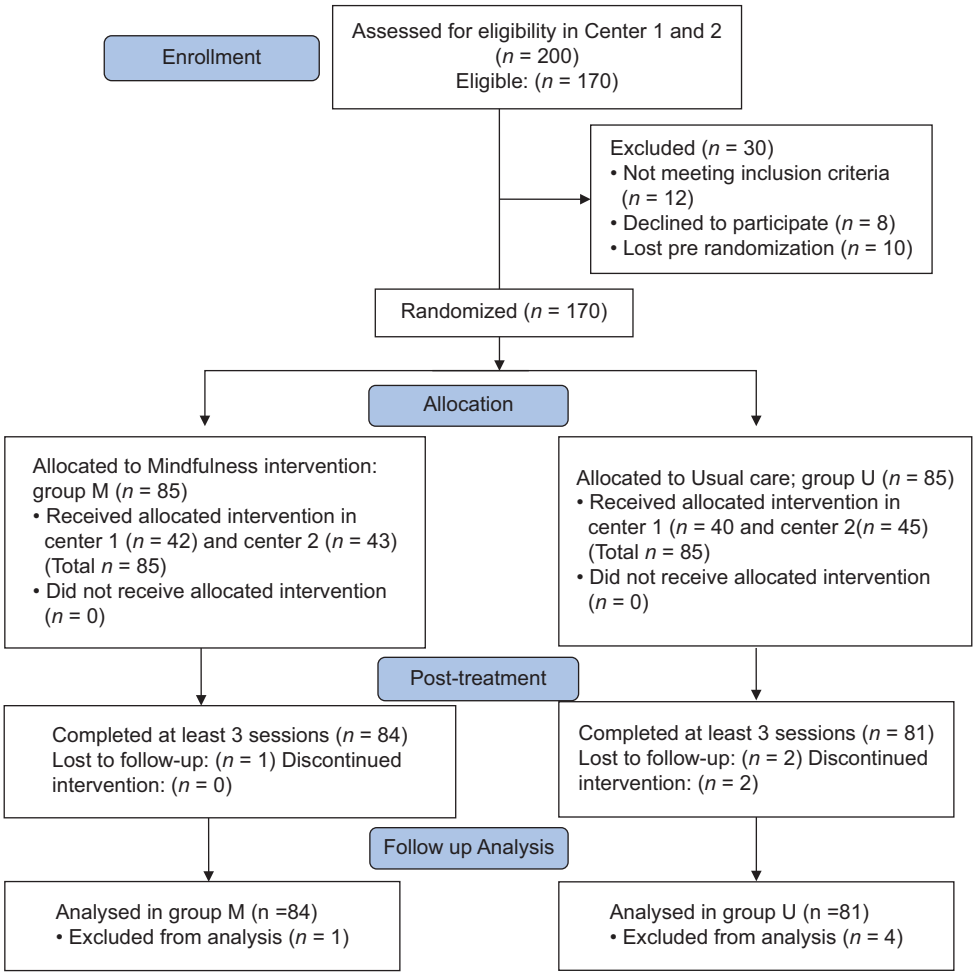



**Figure 1:** Representation of recruitment, study progression, and analysis of participants

**Table 1: An overview of five-session mindfulness program for chronic pain patients**

| Session focus | In-session mindfulness practices |
|---|---|
| Introduction to mindfulness | Mindfulness of breathing |
| Mindfulness and chronic pain | Body scan |
| Application of mindfulness to deal with pain catastrophizing and to increase chronic pain acceptance. | Mindfulness of pain areas |
| Interrupting the cycle of sufferings: moving forward with acceptance and self-compassion. | Self-compassion practice |
| Sum up of all learnings | Review of all practices |

original MBSR program and 'mindfulness for managing pain' program (https://www.ummhealth.org/center-mindfulness) and had an experience of teaching mindfulness for more than 3 years. The regional language was used for communication as most of the participants had low educational levels and a poor understanding of other languages.

The group U participants attended similar sessions by the same instructor. In those sessions, different pain education, self-care, and relaxation techniques (excluding mindfulness concepts) were discussed and adviced to practice daily in a similar schedule like group M. There was a provision for group U participants to join the mindfulness program after the stipulated study period, if they desired.

Attending at least three of five sessions was mandatory to remain in the study. Any participant experiencing any adverse effects was informally asked about exacerbation of anxiety, any experience of psychosis, or exacerbation of pain and was requested to report to the study coordinators.

### Measurement parameters

The baseline variables included demographic parameters like age, sex, religion, education, and provisional diagnosis of pain conditions and were noted before the start of the sessions. The other outcome variables were assessed at two time points: pre- and post-program.

### Pain intensity

The pain intensity was measured using the 11-point NRS. The patients were asked to mark a number between 0 and 10, which correlates best to their pain intensity. Here, 0 represents 'no pain at all', whereas the upper limit represents 'the worst pain ever possible'. NRS (0-10) is a commonly used tool in research studies and has high correlations with other pain-assessment tools.[11]

### Pain catastrophizing

PCS is an instrument for measuring catastrophic thinking related to pain and was used extensively in clinical practice and research. This 13-item questionnaire, each ranging from 0 to 4, enquires about the thoughts and feelings about pain and encompasses three components of catastrophizing: rumination about pain, magnification of the pain, and helplessness to manage the pain. A total PCS score of 30 represents a clinically relevant level of catastrophizing.[12]

### Pain acceptance

An 8-item short-form version of the CPAQ-8 was used in this study. It has two components: activity engagement and pain willingness. In this scale, items 2, 4, 7, and 8 (reverse-scored) look into pain willingness and items 1, 3, 5, and 6 look into activity engagement.[13]

### Perceived stress

PSS-4 is a 4-item stress assessment scale with five responses, and items 2 and 3 are reversed. This scale measures stress by noting feelings and thoughts of participants in the past 1 month. There can be a score of 0 to 16, and higher scores correlate to more stress.[14]

### Well-being

The World Health Organization 5 item well-being index (WHO-5) is a global rating scale measuring subjective psychological well-being. This scale also has adequate validity as a screening tool for depression.[15]

### Mindfulness

FMI is a valid and reliable questionnaire with 14 items covering all the different aspects of mindfulness. The advantages of this scale were that it can be used in subjects without previous meditation experience and it is highly sensitive to change.[16]

### Statistical analysis

The investigator collecting the data and doing analysis was unaware of group allocations. Data were tested for equality of variance using Levene's test. Normality was tested using Shapiro–Wilk test. Baseline characteristics (age, sex, religion, education, and diagnosis) were tested using unpaired *t*-test for age and Chi-square ($\chi 2$) test for other parameters. The outcome parameters were analyzed using repeated measures ANOVA (Bonferroni model). The statistical software used was SPSS Statistics for Windows 7® version 18.0.0 (Chicago, IL 60606-6412), GraphPad Prism® InStat version 5.0 (California 92037-3219), and Microsoft® Office Excel 2010 (Washington: Microsoft). Results were presented as mean (SD) and percentage. $P < 0.05$ was considered statistically significant.

## RESULTS

Baseline characteristics (age, sex, religion, education, and diagnosis) were similar between the cases (group M) compared to the controls (group U) [Table 2].

**Table 2: Representation of demographic characteristics of study participants**

| | Group M (*n*=84) | Group U (*n*=81) | (Group M vs. U) |
|---|---|---|---|
| AGE | | | Unpaired *t*-test |
| Mean Years (SD) | 44.77 (13.24) | 44.62 (13.41) | *P*=0.9400<br>*t*=0.0754<br>df=163, SE (d)=2.075<br>95% CI =-3.94 to 4.25 |
| SEX | | | Chi-square ($\chi^2$) test |
| *n* (%) | MALE=24 (28.57%)<br>FEMALE=60 (71.43%) | MALE=22 (27.16%)<br>FEMALE=59 (72.84%) | *P*=0.8398<br>$\chi^2$=0.0408<br>df=1 |
| RELIGION | | | Chi-square ($\chi^2$) test |
| *n* (%) | MUSLIM=31 (36.90%)<br>HINDU=53 (63.10%) | MUSLIM=31 (38.27%)<br>HINDU=50 (61.73%) | *P*=0.8562<br>$\chi^2$=0.0328<br>df=1 |
| EDUCATION | | | Chi-square ($\chi^2$) test |
| *n* (%) | BLX=25 (29.76%)<br>X-XII=49 (58.33%)<br>G=10 (11.91%) | BLX=23 (28.40%)<br>X-XII=24 (59.26%)<br>G=10 (12.34%) | *P*=0.9806<br>$\chi^2$=0.0391 df=2 |
| DIAGNOSIS | | | Chi-square ($\chi^2$) test |
| *n* (%) | OA=28 (33.33%)<br>LBP=37 (44.05%)<br>M=19 (22.62%) | OA=24 (29.63%)<br>LBP=35 (43.21%)<br>M=22 (27.16%) | *P*=0.7678<br>$\chi^2$=0.5284 df=2 |

In the above table, columns 2 and 3 show age, sex, religion, education, and diagnosis distribution for group M (case group) and group U (control group). Column 4 shows the *P* value of unpaired *t*-test/$\chi^2$ test comparison prior to beginning of session. *n*=Number of participants in the group, M=Mindfulness group, U=Usual care group, BLX=Education below Class X, X-XII=Education level X to XII, G=Education level graduate, OA=Osteoarthritis, LBP=Low-back pain, M=mMusculoskeletal pain (non-specific)

Regarding the outcome parameters [Table 3], results of repeated measures ANOVA (RM-ANOVA) show a significant difference [F(1,326) = 15.0122; *P* = 0.0001291] between pre-session and post-session in NRS (pain intensity). The treatment effect size was $\eta_p^2$ = 0.04402. Similar statistically significant differences were observed for pain acceptance [F(1,326) = 4.5311; *P* = 0.03403] and perceived stress score [F(1,326) =13.2788; *P* = 0.0003122]. The treatment effect size was $\eta_p^2$ = 0.01371 and $\eta_p^2$ = 0.03914, respectively. Variance comparison between the case (group M) and control (group U) shows a significant difference in pain acceptance [F(1,326) = 8.8186; *P* = 0.003203], pain catastrophizing [F(1,326) = 8.0009; *P* = 0.004965], PSS [F(1,326) = 9.1501; *P* = 0.002685], and FMI [F(1,326) =11.9392; *P* = 0.0006224]. No significant difference between pre-session and post-session was observed for PCS [F(1,326) = 1.7762; *P* = 0.1835], WHO-well-being scale [F(1,326) = 1.3821; *P* = 0.2406], and FMI [F(1,326) = 1.0625; *P* = 0.3034]. However, further studies with a larger sample size along with provision of the follow-up period will be needed to substantiate the results (outliners of the above test were done by Tukey Fence with k = 1.5).

No adverse effect was reported by any participants.

## DISCUSSION

The results approved the hypothesis formed. Some demographic parameters noted can serve as an interesting database in Indian context. Around 36.9–38.2% participants were Muslims and the rest Hindus: both had no objections/taboos to the teachings and practices of mindfulness (which were claimed to be secular). Only 11.9–12.3% of participants had an education level of graduation or above, but no participants had any difficulties in understanding and execution (the regional language was used in most parts). Low formal education does not seem to be a barrier in learning mindfulness. Regarding the diagnosed pain conditions in different groups, 29.6–33.3% patients were suffering from osteoarthritis, 43.2–44% from low back pain, and the rest from non-specific musculoskeletal pain. The mindfulness group participants showed some significant changes in outcome variables post program, when compared to control groups.

### Pain intensity

An RCT with 109 patients with non-specific chronic pain found small (non-significant) changes for pain intensity.[17] A review of literature found 10 out of 16 studies on mindfulness, reporting a decrease in pain intensity for chronic pain patients,[18] while in our subjects, a significant decrease in pain intensity with NRS [F(1,326) =15.0122; *P* = 0.0001291] between pre-session and post-session was noted, which was comparable with previous studies.

The psycho-social variables influencing pain are complex and diverse, affecting multiple areas of functioning – cognition, emotion, and behavior. The psychological and social factors are not secondary reactions to persistent pain; rather, they form an interactive complex that characterizes chronic pain itself. In this study, we have only focused on three most important psycho-social factors: pain catastrophizing, fear of avoidance of work (reflected by pain acceptance: pain willingness and activity engagement), and low expectations for recovery and life (reflected by well-being).[5] The other complex aspects of the psychosocial dimension were beyond the scope of this study.

### Pain catastrophizing

Catastrophizing is a cognitive and emotional response to pain consisting of magnification of pain sensations (e.g., "This is the worst pain I've ever experienced"), rumination about pain and pain-related concerns (e.g., "I can't stop thinking about the pain"), and helplessness about one's ability to manage pain. Pain catastrophizing (an exaggerated negative response

**Table 3: Representation of outcome variables in mindfulness (group M) and usual care (group U) using RM-ANOVA**

| Variable | Session | Group M ($n$=84) | Group U ($n$=81) | Comparison | Mean square (MS) | $F$ statistic ($DF_1$, $DF_2$) | $P$ | Effect size ($\eta^2$) |
|---|---|---|---|---|---|---|---|---|
| NRS: Pain intensity | Pre-session | 6.69 (1.22) | 6.44 (0.74) | Between cases (group M) and usual care (group U) | 1.6316 | 1.8006 (1,326) | 0.1806 | $\eta^2$=0.044 |
| Mean (SD) | Post-session | 5.90 (0.98) | 6.43 (0.67) | Between pre-session and post-session | 13.603 | 15.0122* (1,326) | 0.0001291** | Treatment effect size $\eta_p^2$=0.04402 |
| CPAQ: Pain Acceptance | Pre-session | 20.94 (3.24) | 20.60 (3.53) | Between cases (Group M) and usual care (Group U) | 105.8987 | 8.8186* (1,326) | 0.003203** | $\eta^2$=0.013 |
| Mean (SD) | Post-session | 22.54 (3.38) | 20.60 (3.61) | Between pre-session and post-session | 54.4121 | 4.5311* (1,326) | 0.03403** | Treatment effect size $\eta_p^2$=0.01371 |
| PCS: Pain Catastrophizing | Pre-session | 35.85 (9.79) | 37.41 (9.95) | Between cases (group M) and usual care (group U) | 741.055 | 8.0009* (1,326) | 0.004965** | $\eta^2$=0.0053 |
| Mean (SD) | Post-session | 33.02 (9.11) | 37.46 (9.52) | Between pre-session and post-session | 164.5121 | 1.7762 (1,326) | 0.1835 | Treatment effect size $\eta_p^2$=0.005419 |
| WHO-Well-being Mean (SD) | Pre-session | 9.74 (3.59) | 9.64 (3.45) | Between cases (group M) and usual care (group U) | 25.394 | 2.1734 (1,326) | 0.1414 | $\eta^2$=0.0042 |
| | Post-session | 10.63 (3.24) | 9.62 (3.36) | Between pre-session and post-session | 16.1485 | 1.3821 (1,326) | 0.2406 | Treatment effect size $\eta_p^2$=0.004222 |
| PSS: Perceived Stress Scale | Pre-session | 8.40 (1.48) | 8.32 (1.41) | Between cases (group M) and usual care (group U) | 17.2917 | 9.1501* (1,326) | 0.002685** | $\eta^2$=0.038 |
| Mean (SD) | Post-session | 7.32 (1.20) | 8.32 (1.27) | Between pre-session and post-session | 25.0939 | 13.2788* (1,326) | 0.0003122** | Treatment effect size $\eta_p^2$=0.03914 |
| FMI: Mindfulness characteristics | Pre-session | 23.05 (5.18) | 21.77 (5.21) | Between cases (group M) and usual care (group U) | 313.8011 | 11.9392* (1,326) | 0.0006224** | $\eta^2$=0.0031 |
| ean (SD) | Post-session | 24.29 (5.18) | 21.67 (4.88) | Between pre-session and post-session | 27.9273 | 1.0625 (1,326) | 0.3034 | Treatment effect size $\eta_p^2$=0.003249 |

In this table, Column 1 describes the outcome variables, Column 2 describes the session status (viz. Pre-session or Post-session) beside their corresponding mean and standard deviations, Column 3 and 4 describe the outcome variables (mean (SD)) for the case group (group M) and control group (group U), respectively, Column 5 describes the comparison between groups and sessions using the RM-ANOVA, Column 6 shows intermediate calculations of mean squares, Columns 7, 8, and 9 describe the F-statistic, *P*-value, and effect size, respectively. ANOVA=Analysis of variance, NRS=Numerical Rating Scale, PSS=Perceived Stress Scale, FMI=Freiburg mindfulness inventory, DF=Degrees of freedom, SD=Standard deviation, MS=Mean Square, $\eta^2$=Effect Size of the difference of variance by RM-ANOVA, $\eta_p^2$=Treatment effect size. **F*-Statistic is statistically significant, ***P*<0.05 are statistically significant

to pain) represents an important process factor in pain treatment.[19] There is a dearth of prevalence data on pain catastrophizing in Indian context. But a study doing validation of the Hindi version of PCS in chronic low-back pain patients can be indirect evidence of its significant prevalence in Indian patients.[20] A previous study found MBIs for chronic pain to be effective via reductions in pain catastrophizing.[21] In our study, there was an impact on pain catastrophizing in the mindfulness group [F(1,326) = 8.0009; $P$ = 0.004965] compared to control group.

### Pain acceptance

Pain acceptance helps a person to reduce unsuccessful attempts to avoid/control pain and focus on participation in valued activities and personally relevant goals. It is "a willingness to experience continuing pain without needing to reduce, avoid, or otherwise change it."[22] Pain acceptance leads to enhanced emotional/physical functioning in chronic pain patients beyond the influence of depression, pain intensity, and coping.[23] A previous Western study on 107 participants found mindfulness to heighten pain acceptance (d = 0.66),[24] while another with 109 patients showed similar results.[17] In our study, chronic pain acceptance improved post session [F(1,326)= 4.5311; $P$ = 0.03403] and compared to control group [F(1,326) = 8.8186; $P$ = 0.003203]. The results were comparable to previous studies.

### Perceived stress

There were no available baseline data about prevalence of perceived stress in chronic pain patients. Higher perceived stress was independently associated with low back pain.[25] Pain theories emphasize the importance of psychological and stress-related factors in the perception and persistence of pain. Higher self-perceived stress reactivity is known to be associated with increased chronic pain risk. A previous study by Omidi *et al.* found mindfulness to decrease perceived stress in tension headache patients in Iran.[26] A study by present researchers found mindfulness to decrease perceived stress in the general Indian population (non-pain patients).[27] In this study, mindfulness made a significant decrease in perceived stress [F(1,326) = 13.2788; $P$ = 0.0003122] of the chronic pain patients post session comparable to studies done in the past.

### Well-being and mindfulness

Positive psychological well-being predicts lower severe pain in the general population.[28] A study found mindfulness to improve mental well-being and quality of life in healthy Indian adults.[29] In our study, the effects on WHO well-being [F(1,326) = 1.3821; $P$ = 0.2406] and mindfulness characteristics measured by FMI [F(1,326) = 1.0625; $P$ = 0.3034] were non-specific. Minimal changes were expected in these parameters considering the brief period of intervention. A longer follow-up will be perhaps necessary to detect changes.

The results of our study corroborate with the previous research in other parts of the world. Presently, there was no research available to provide baseline data about the measured outcome variables in Indian chronic pain patients. Anxiety and depression are two common comorbidities highly prevalent with chronic pain conditions.[30] The present study does not look specifically into these comorbidities (patients with psychiatric illness were excluded); rather, it focuses on coexisting psycho-social associations of chronic pain. The authors are of an opinion that a chronic pain patient with psychiatric illness should be treated individually by mental health experts.

In the western world, research with mindfulness has come a long way. Mindfulness was found to improve pain treatment outcomes and potentially decrease pain-related medication utilization.[31] A systemic review and meta-analysis found mindfulness to improve pain.[9] The American college of physicians included mindfulness in their clinical practice guidelines for chronic low back pain.[32] A study by Zeidan *et al.* outlined the detailed mechanisms behind mindfulness-based pain relief.[8] Specific mindfulness-based pain management programs have been developed and researched upon.[33]

In Indian context, the research with mindfulness in chronic pain patients remains minimal. Patil *et al.* only mentions its use in chronic low back pain.[34] A pilot study from Chennai explores the effect of mindfulness on well-being.[35] A small study by present investigators saw potential in mindfulness affecting different variables in Indian chronic pain sufferers.[10]

The major strengths of this study are the bi-centric format and a considerable sample size. Perhaps for the first time, the psycho-social variables affecting chronic pain were brought to focus in Indian context. It provides an important database upon which more intensive research work can take place. The true secular nature of mindfulness got re-established by participation from both religious communities. A low educational level was not a barrier to learning of mindfulness (suitable for masses). A brief format (five instead of eight sessions) appears to be useful for participant retention (dropouts were negligible). Arranging dedicated places for sessions in busy government hospitals was a bit of a challenge (but manageable). The provision to interact and share their feeling/sufferings during sessions appears very beneficial for the participants. A low rate of dropouts can be attributed to dedicated work of the institutional caregivers and extensive effort of study co-ordinators.

### Limitations

All the outcome measures were subjective, self-reported by the participants. The interest of patients to participate may have influenced the results, and expectancy bias cannot be ruled out. Physical attendance was challenging for some patients with severe pain. For home practice, researchers depended on verbal confirmation of patients. Intent to treat analysis could not be done and per protocol analysis was conducted, though the dropouts in the study were minimal. No comments can be made on cost of intervention as cost analysis was not performed. The exploration of other diverse psycho-social variables was beyond the scope of this study. Follow-up data collection to study long-term effects of mindfulness was not possible.

Despite the limitations, this study is a stepping stone in addressing the psycho-social components of chronic pain in Indian context.

## CONCLUSION

Mindfulness was found to decrease pain intensity, decrease pain catastrophizing, and increase chronic pain acceptance helping Indian patients to cope better with chronic pain. There was a positive impact on perceived stress, mindfulness, and non-specific changes in well-being. Further studies will be required to substantiate these results.

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### Conflicts of interest

There are no conflicts of interest.

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